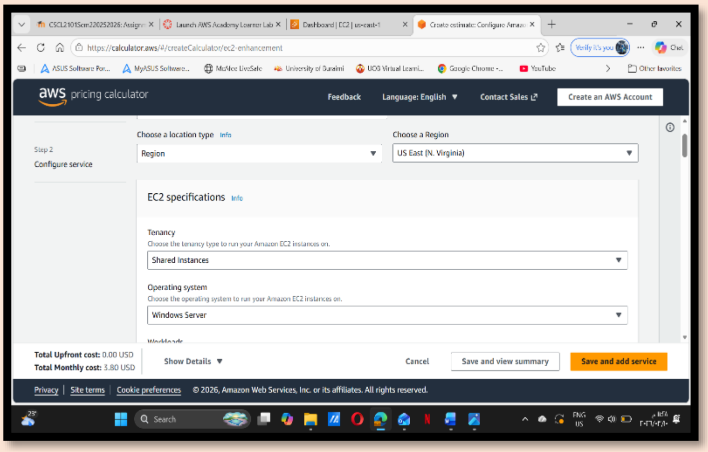
Task1:

CLOUD COST OPTIMIZER



Using the AWS Pricing Calculator, the monthly cost for the cloud infrastructure was estimated in the US East (N. Virgini) region. The configuration includes a Windows Server instance, with an estimated total monthly cost of $3.80 USD.

1. INTRODUCTION:

The objective of this project is to demonstrate the deployment of a functional web application on a cloud environment. Using Amazon Web Services (AWS), a Windows Server instance was configured to host a PHP application connected to a MySQL database.

2. Tools& Technologies:

PURPOSE	TOOL
Cloud Infrastructure (Windows Server 2022)	AWS EC2
Local Server Stack (Apache & MySQL)	XAMPP
Back-end Scripting Language	PHP
Database Management Interface	phpMyAdmin

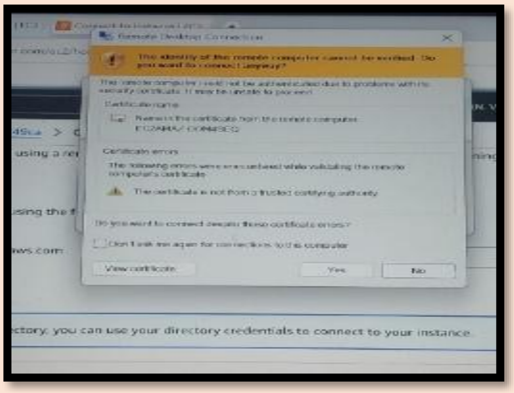
3. Implementation Steps:

Step1:

Accessing the Cloud Instance.

- **Image:**

Connection via Remote Desktop using the Public DNS.

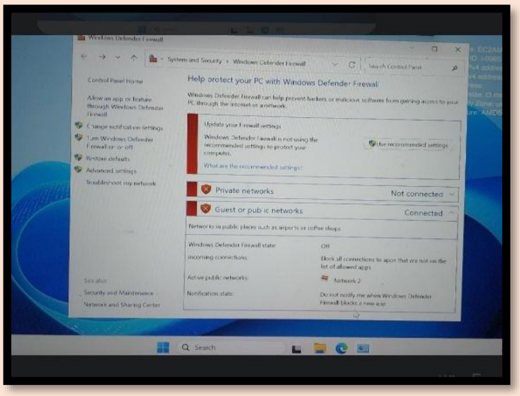


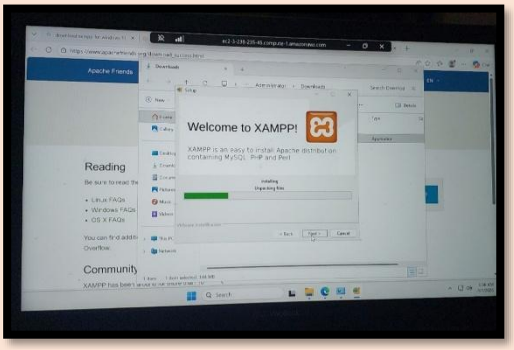
Step2:

Server Configuration (Firewall & XAMPP).

- **Image:**

Disabling Windows Firewall.



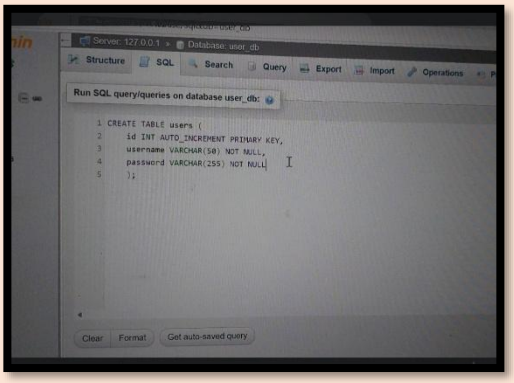


- **Image:**

  Installation of XAMPP.

- **Description:**

  Firewall settings were adjusted to allow HTTP traffic, and XAMPP was installed to provide the necessary web services.

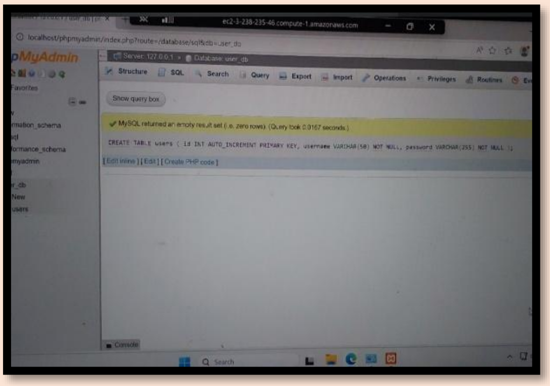


Step3:

Database & Table Creation

- **Image***:*

  Creation of the Database & Table



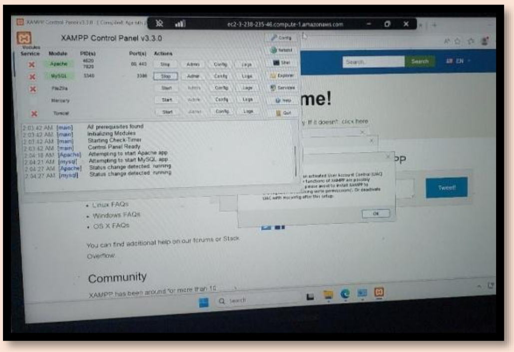
- **Image:**

  SQL script execution for the users table.

- **Description:**

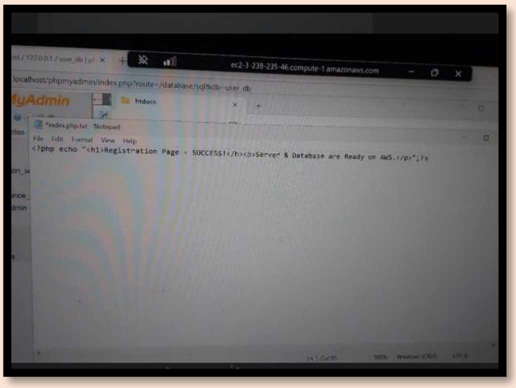
  A structured database was created to store user information, defining fields for ID, Username, and Password.

Step4: Application Development (PHP)



- **Image:**

  Writing the index.php code in Notepad.



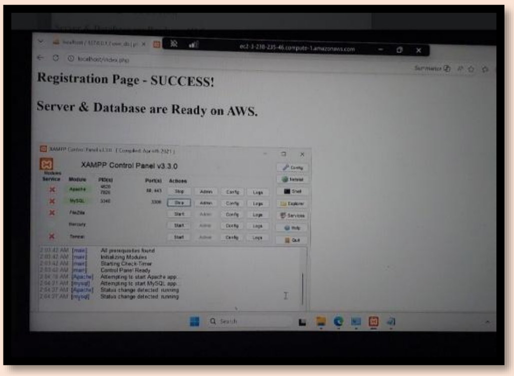
- **Image:**

  Saving the file in the htdocs directory.

- **Description:**

  A PHP script was developed to test the integration between the web front-end and the MySQL back-end.

4. Final Results & Verification:



- **Image:**
  Browser success message and XAMPP status.

- **Description:**

The final verification shows the website running successfully on the AWS Cloud. The success message "Registration Page – SUCCESS!" confirms that all components are working together perfectly.

5. Conclusion:

This project successfully illustrates the power of cloud computing in hosting scalable applications. By integrating AWS EC2 with the XAMPP stack, we achieved a fully operational web environment ready for production.